# Canary SSM: Implementation Brief for Antigravity (rev. 2)

Companion to the repository **canary_ssm_project**. Revision 2 folds in two handoff documents summarising pilot experiments 1-6 (second-hand: the pilot code and CSVs were not seen) and my own null audit of this repo. It makes **no performance claims**. Status per the handoffs and my audit: **not GO**. Calibration is the blocker.

## 1. What changed since revision 1

- **Bug fixed:** null rollouts were labelled `kind="null"`, which pandas reads as missing, so null-rate tables were silently empty. Now `no_event`, with a regression test and a hard error on empty null tables.

- **Noise model:** default is a single AR(1) per host (white SD pinned); AR+white and white-only are ablations so the missing comparison table is produced.

- **Scenarios:** sustained workload step (log 1.4), class-wide harm, 30% class-label errors, noise sweep 0.01/0.02/0.04, phi_e and heavy-tail stress.

- **Adjustment:** `--beta-mode pre|pooled_dev`. **Effect grid:** long-run SD of the effect state must be <= kappa (24 -> 18 candidates). **Huber** ablation.

- **New tools:** `audit-null`, frozen fit (`--frozen`), joint-calibrated combined detectors, numeric PASS/FAIL gate (`gate_status.json`), abstention table.

- **Tests:** 41 (was 30). All pass in the sandbox.

## 2. What the pilot experiments and my audit establish

| Finding | Trust | Why |
|---|---|---|
| Kalman filter correct, causal, handles missing data; runtime far below 1 s | Established | Scalar fixture, naive reference, statsmodels (pilot); same checks pass in this repo's tests |
| Naive pre/post fails under a shared workload step | Established | 100% false alerts in pilot |
| Adjusted detectors keep 5% under a shared step | **Not met** | Pilot D1-D3 11.7/59/61%; reproduced here (table below) |
| Calibration on step-contaminated nulls inflated a CUSUM threshold about 2x (not 16x) | Established | Resolved the CUSUM contradiction between pilot experiments |
| SSM release scores are calibrated by the order-statistic rule | **Contradicted** | Pilot: 42/500 = 8.4% [6.1, 11.2] even with true parameters; my plain-null audit is near target at small n, so audit at n >= 400 |
| Hierarchy is best overall | **Not supported** | Wins class-wide harm; loses sparse harm; ties/loses broad; unequal false-alert rates |
| Persistence helps | Supported, one noise level | No-persistence slower by about 34 min on S1 (pilot), calibration caveat applies |
| Single AR(1) is the right noise model | Plausible, unsupported | No comparison table in the pilot; one contrast points the other way. Pipeline now produces the table |

**My null audit (generated from docs/sandbox_audit CSVs).** Frozen declared hyperparameters (not tuned), thresholds calibrated on 300 plain nulls, then 200 fresh nulls per family; 95% Clopper-Pearson intervals in brackets (about +/- 5 points, so only large gaps mean anything). Target is 5%.

| method / beta mode | plain | step (homog.) | step (heterog. beta) | phi_e 0.8 | Student-t(3) |
|---|---|---|---|---|---|
| matched_contrast / pre | 4.0% [2,8] | 12.0% [8,17] | 8.0% [5,13] | 3.5% [1,7] | 22.0% [16,28] |
| adj_cusum / pre | 8.5% [5,13] | 68.0% [61,74] | 68.0% [61,74] | 33.5% [27,40] | 7.0% [4,11] |

| method / beta mode | plain | step (homog.) | step (heterog. beta) | phi_e 0.8 | Student-t(3) |
|---|---|---|---|---|---|
| topk_cusum / pre | 6.0% [3,10] | 67.0% [60,73] | 65.0% [58,72] | 34.5% [28,42] | 6.0% [3,10] |
| hier_ssm / pre | 6.5% [4,11] | 18.0% [13,24] | 16.0% [11,22] | 26.0% [20,33] | 11.5% [7,17] |
| matched_contrast / pooled_dev | 3.5% [1,7] | 10.5% [7,16] | 18.0% [13,24] | 3.0% [1,6] | 20.0% [15,26] |
| adj_cusum / pooled_dev | 4.5% [2,8] | 39.0% [32,46] | 92.0% [87,95] | 31.0% [25,38] | 6.5% [4,11] |
| topk_cusum / pooled_dev | 4.5% [2,8] | 44.5% [37,52] | 93.5% [89,96] | 39.0% [32,46] | 6.0% [3,10] |
| hier_ssm / pooled_dev | 5.5% [3,10] | 7.5% [4,12] | 43.5% [37,51] | 27.0% [21,34] | 11.5% [7,17] |

Reading: the pilot's shared-step failure reproduces here. A development-pooled slope helps when hosts respond alike and fails badly when they do not, so neither adjustment mode is safe. Persistent noise (phi_e 0.8) inflates every model-based detector; heavy tails inflate matched contrast. Nothing here says which detector is faster; frozen values audit calibration only.

## 3. Setup (Windows)

```
py -3.11 -m venv .venv
.venv\Scripts\activate
python -m pip install -e ".[dev]"
python -m pytest -q                # expect: 41 passed
```

CPU only (the handoffs measured the filter at about 0.1 ms per update; the RTX 3060 is unused). Put AGENTS.md in the workspace rules if your Antigravity build loads them; otherwise paste its hard rules into every new session (I cannot verify how your version loads rule files).

## 4. Hard rules (AGENTS.md)

- Detector modules never import Truth or metrics, and never read *_truth files (test enforced). Causal scores only; per-host fits use the pre-period.

- Roles never mix: dev fits/selects, calibration sets thresholds, null_validation audits, test is final. Calibrate on the distribution you evaluate on.

- PASS/FAIL comes from numbers. 8.4% [6.1, 11.2] is a FAIL. The gate needs n >= 400 nulls per cell; fewer is reported NOT_EVALUABLE, never PASS.

- Report SD/RMS of per-rollout post-minus-pre residual shifts, not signed means (a signed mean cannot fail).

- Choose K, thresholds and hyperparameters on development data only, never from the true affected count. Compare delay only among methods inside the gate.

- No claims of causal attribution, calibration validated, hierarchy beats classical baselines, real-telemetry value, or novelty.

## 5. Run order

```
bash scripts/smoke.sh            # plumbing, ~4 min, no accuracy reported
bash scripts/audit_quick.sh      # frozen calibration audit incl. beta modes, ~15-25 min
.\scripts\run_all.ps1            # full benchmark + audit-null (500/cell) + noise sweep
# stage by stage: simulate -> fit -> calibrate -> validate-null -> audit-null -> evaluate (reads gate_status.json)
python -m canary_ssm.cli audit-null --config configs/audit.yaml --model artifacts/model \
    --policy artifacts/policy --n 500
```

Read `null_validation.json`, `results/audit/table_audit_null.csv` and `gate_status.json` before any detection table. Fit time estimate: about 18 SSM candidates x 3 families on 600 dev rollouts, tens of minutes on one CPU core (estimate from sandbox timing; run a 20-pair dry run first).

## 6. Definitions that must not change silently

| Item | Definition |
|---|---|
| Harm / event | kappa = log(1.02); effect equal to kappa is non-harmful; qualifying event = run of >= 15 minutes on >= 1 canary; alarms latch per rollout |
| Primary metric | Horizon-capped time to success (misses and premature alarms count as the remaining horizon); also report miss, premature, late-after-recovery rates |
| Threshold | k = ceil((B+1)(1-alpha)); alert when a later full-horizon maximum strictly exceeds the k-th smallest of B null maxima (B=500: k=476) |
| Release score (SSM) | log max over hosts of Pr(a_i > kappa): a score, not Pr(any host harmful). 3% latency = log(1.03) = 0.0296, so an average over 3 of 10 canaries (about 0.0089) is far below kappa |
| Effect grid | rho in {0,.8,.95,.99} x innovation SD {.001,.005,.01} x prior SD {.01,.03}, keeping only long-run SD sd/sqrt(1-rho^2) <= kappa |
| Combined detector | score = max over components of each score's rank among calibration-null maxima; one joint threshold (a naive OR of separate thresholds exceeds the budget) |

# 7. Next experiments (outline; each needs its own plan and estimate)

| # | Question | Tools in repo / what is missing |
|---|---|---|
| 7 | Can calibration hold under a shared step, and can a CAUSAL rule pick the calibration regime from the observed workload shift? | audit-null, configs/audit.yaml exist. Regime rule is NOT implemented: test rule on step and plain nulls, false-alert rates must hold under the rule itself |
| 8 | Does adjustment survive heterogeneous host slopes, phi_e misspecification and heavy tails? | beta_jitter, phi_e, student_t3 families and hier_huber exist. Need robustness fixes and per-rollout residual-spread diagnostic |
| 9 | At validated equal false-alert rates, who wins by scenario and noise level, including combined detectors? | combo_* detectors, noise_sweep.yaml, paired bootstrap exist. Need S6, A2 reporting, pooled-over-scenario contrasts as pre-registered |

# 8. Copy-ready prompts

One at a time. Require: files changed, commands run, pytest output, and no edits to existing tests.

## A. First run and review

Read AGENTS.md and docs/DESIGN_DECISIONS.md. Run the tests, then scripts/smoke.sh. Report each stage, the null alert table with exact intervals, gate_status.json (it should say NOT_EVALUABLE at smoke size), fit_failures.csv and warnings. Do not change code; list anything suspicious with evidence.

## B. Calibration audit at size

Run scripts/audit_quick.sh, then audit-null with n=500 on the full configs/audit.yaml. Produce results/audit/summary.md via scripts/summarize_audit.py. For every cell above 5% name the likely mechanism using saved artifacts only; do not change thresholds or hyperparameters.

## C. Experiment 7: causal regime rule

Propose, in writing first, a causal rule that selects the calibration regime from the measured workload covariate only (no truth, no post-hoc labels). After I approve, implement it, calibrate each regime on its own nulls, and show false-alert rates on step and plain nulls under the rule at n >= 400 per cell. Report the power cost against single-threshold calibration.

## D. Experiment 8: robustness

Evaluate matched contrast, pooled beta and per-host beta under heterogeneous slopes; compute SD/RMS of per-rollout post-minus-pre residual shifts and the spread of beta-hat about true beta per variant. Evaluate hier_huber and a conservative phi_e on phi_e=0.8 and Student-t(3). Report power alongside false alerts.

### E. Experiment 9: benchmark

After the calibration gate passes for the methods in scope, run the equal-false-alert benchmark on S1-S5 at noise 0.01/0.02/0.04, including A1-A4 ablations and combo_* detectors. Pre-registered contrasts must be computed per scenario and per noise level and pooled as registered. Write the report only through generated tables.

### F. Independent audit

Audit for: truth leakage, test reuse, calibration/validation role mixing, tie handling and strict exceedance, eligibility handling, AR noise handling, unjustified PASS labels, summary-vs-table contradictions. Return defects with file and line evidence; do not fix in the same step.

## 9. What may and may not be claimed

**May say (declared simulation only):** the filter is numerically verified; the naive pre/post detector fails under a sustained shared step; calibration on contaminated nulls inflates a CUSUM threshold; persistence matters; in class-wide harm the hierarchy was faster than independent-host and pooled SSMs in the pilot.

**May not say:** calibration is validated; the hierarchy beats classical baselines or is generally best (it is slow in sparse harm); the SSM is robust to misspecification; matched contrast is robust outside uniform workload sensitivity; there is a GO decision; results transfer to real telemetry; causal attribution; novelty. A negative result is a valid outcome.

**Not implemented:** a causal runtime regime rule, online beta from controls (confounded with the shared factor shock), Xie-Siegmund and BOCPD comparators, Monte Carlo release probability, rank > 1 factor, informative missingness, real data, dashboard, neural track. Choices where I read the handbook one way are in docs/DESIGN_DECISIONS.md.